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BASIC BLOCK

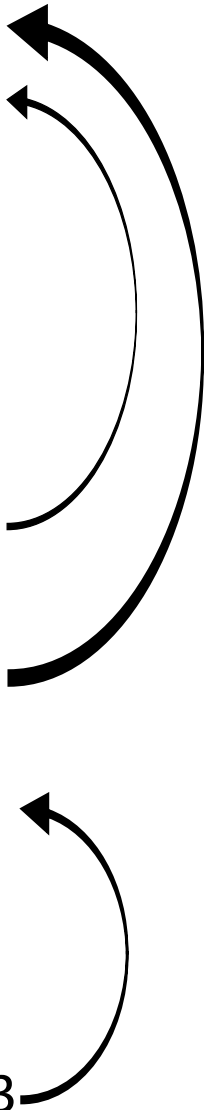
How to select LEADER

1. The first three address instruction in the intermediate code is a leader.
2. Any instruction that is the target of a conditional or unconditional jump is a leader.
3. Any instruction that immediately follows of a conditional or unconditional jump is a leader.

Consider this example to find leader &
divide into blocks:

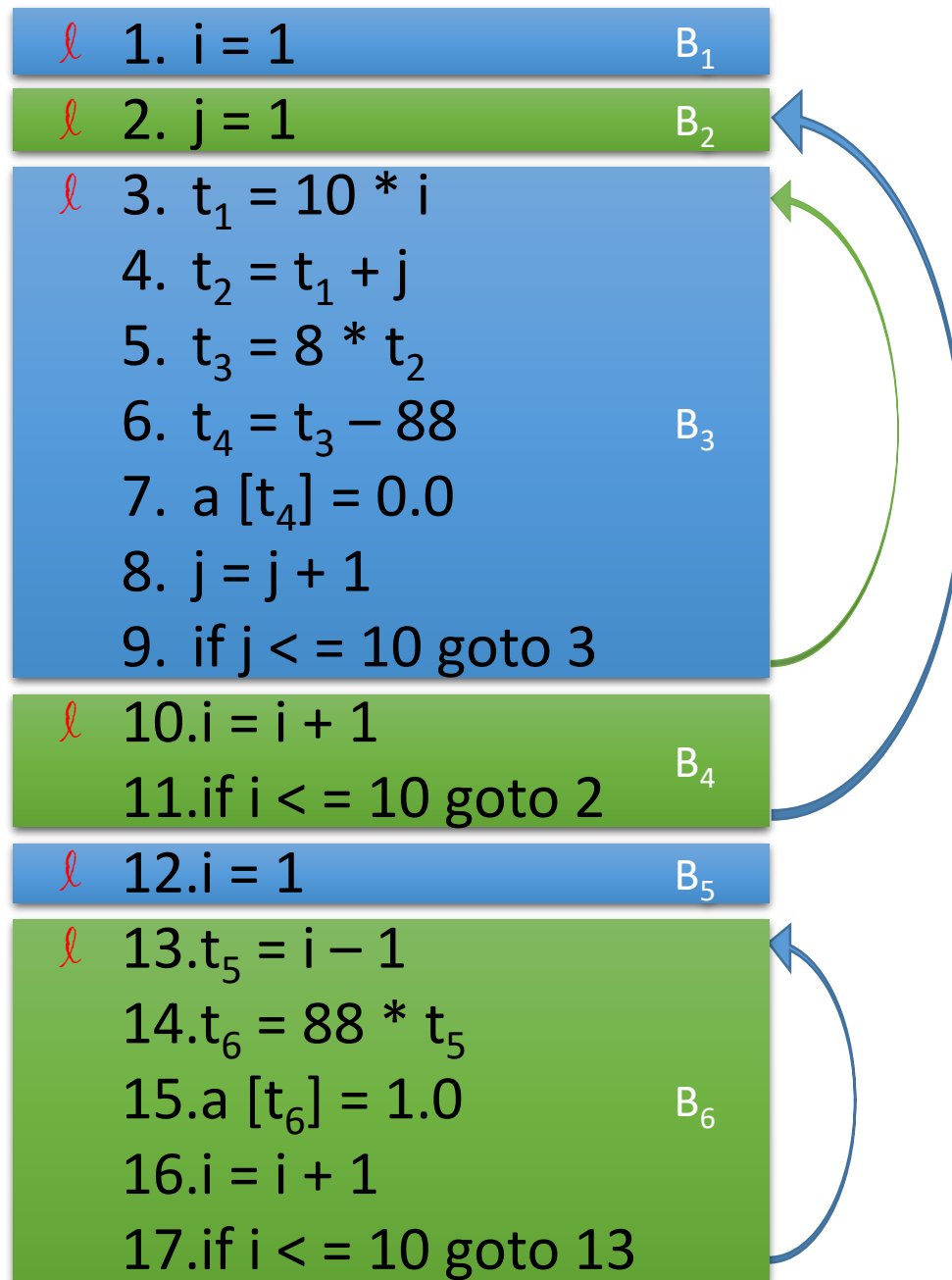
1. $i = 1$
2. $j = 1$
3. $t_1 = 10 * i$
4. $t_2 = t_1 + j$
5. $t_3 = 8 * t_2$
6. $t_4 = t_3 - 88$
7. $a[t_4] = 0.0$
8. $j = j + 1$
9. if $j \leq 10$ goto 3
10. $i = i + 1$
11. if $i \leq 10$ goto 2
12. $i = 1$
13. $t_5 = i - 1$
14. $t_6 = 88 * t_5$
15. $a[t_6] = 1.0$
16. $i = i + 1$
17. if $i \leq 10$ goto 13

ℓ 1. $i = 1$
ℓ 2. $j = 1$
ℓ 3. $t_1 = 10 * i$
4. $t_2 = t_1 + j$
5. $t_3 = 8 * t_2$
6. $t_4 = t_3 - 88$
7. $a[t_4] = 0.0$
8. $j = j + 1$
9. if $j \leq 10$ goto 3
ℓ 10. $i = i + 1$
11. if $i \leq 10$ goto 2
ℓ 12. $i = 1$
ℓ 13. $t_5 = i - 1$
14. $t_6 = 88 * t_5$
15. $a[t_6] = 1.0$
16. $i = i + 1$
17. if $i \leq 10$ goto 13



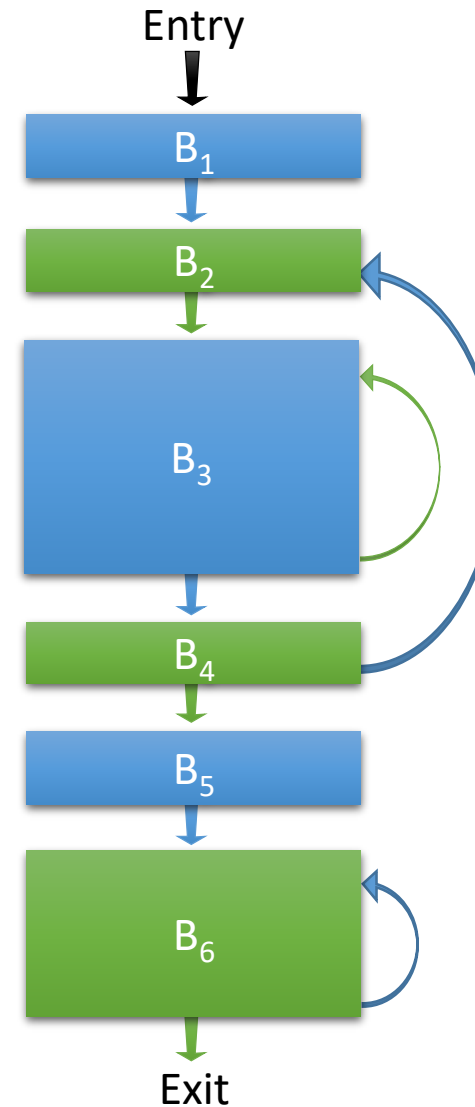
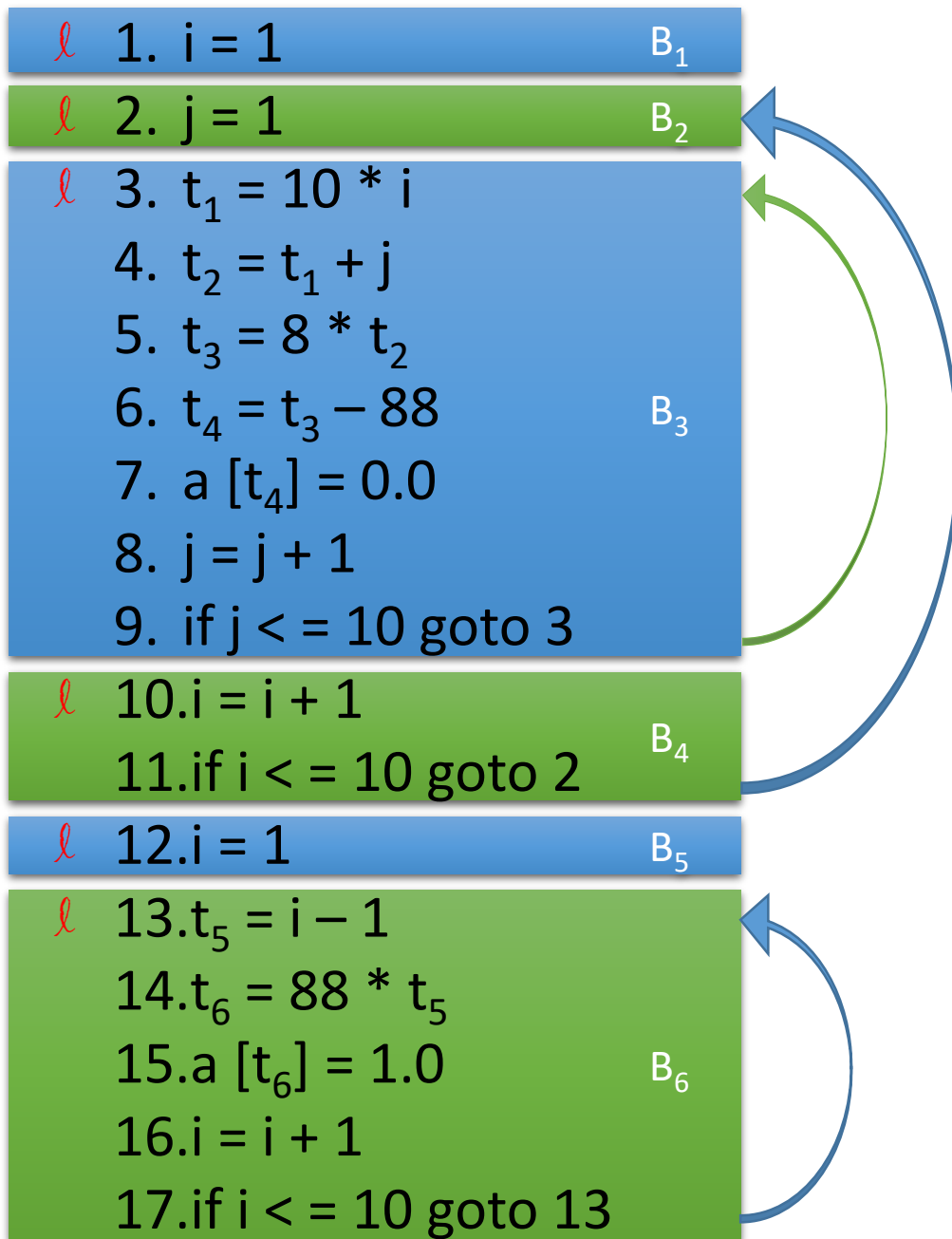
First, we find leader (*ℓ*)

- The first three address instruction in the intermediate code is a leader
- Any instruction that is the target of a conditional or unconditional jump is a leader
- Any instruction that immediately follows of a conditional or unconditional jump is a leader.



First, we find leader (ℓ),
then divide into blocks(B)

Flow Graph



Thank You